ONLINE TRAFFIC FINE PAYMENT SYSTEM (OTFPS)

Submitted by

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DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

CERTIFICATE

This is to certify that the mini project report entitled “Online Traffic Fine Management System” submitted by Ragyam Sinha of B.Tech (CSE), Techno Main Salt Lake, has been carried out under my supervision and guidance. The work embodied in this report is original and has not been submitted elsewhere.

Guide: ……
(Signature)
Name: _____Prof. Poulami Dutta _____
Designation: ____Asst. Professor_____

ACKNOWLEDGEMENT

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Place: Techno Main, Salt Lake

Date:

……

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ABSTRACT

This project report presents the development of a real-life application problem using software engineering principles. It outlines the motivation, methodology, implementation, and outcomes achieved.

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ONLINE TRAFFIC FINE PAYMENT SYSTEM

1. Introduction

1.1. Overview of the problem domain

Traffic fine management in India is still partially manual, causing delays, errors, and lack of transparency. Citizens often need to visit traffic offices, stand in queues, and manually verify fine details. Lack of digitization leads to:

- Loss of records
- Delayed payments
- Manual errors
- Fraudulent entries
- Difficulty verifying vehicle/fine information

The **Online Traffic Fine Payment System (JURMANA)** digitizes this end-to-end process.

1.2. Motivation for the project

The motivation behind developing JURMANA includes:

- Reducing paperwork and manual dependency
- Ensuring transparency between citizens and traffic authorities
- Making fine payment convenient and available 24×7
- Integrating with Parivahan, payment gateway, SMS/Email systems
- Supporting Digital India initiatives
- Minimizing corruption and human intervention

1.3. Scope and objectives

Scope

The JURMANA system covers:

- Online fine viewing
- Real-time payment through gateway
- Automated notifications (SMS/Email)
- Dispute management
- Admin dashboards & reporting
- Integration with external APIs

Objectives

- Provide a secure, user-friendly online platform for citizens
- Automate fine issuance and payment processes
- Enable admin-level monitoring, tracking, and analytics
- Maintain accurate records of fines, payments, disputes, and notifications
- Ensure fast, safe, and verifiable transactions

1.4. Relevance to real-life

The project directly benefits:

- Citizens (easy fine access & payment)
- Police departments (faster fine issuance & tracking)
- Government (digitization & transparency)

This system represents a practical, real-world application used nationwide.

2. Literature Review

2.1. Summary of existing systems or related works

Existing platforms like **Parivahan e-Challan** or city-wise portals provide basic fine viewing, but often lack:

- Real-time sync
- Dispute modules
- Admin dashboards
- Multi-channel notifications
- Local traffic department integration

2.2. Limitations of current approaches

- Slow updates (delays in reflecting fines)
- Limited support for evidence/proof upload
- No transparent dispute resolution
- Inconsistent user interfaces
- Limited analytics for admin
- Manual record-keeping in many regions

2.3. Research gap identification

The gap identified:

- Need for an integrated, state-level system
- Need for automated notifications and payment confirmations
- Lack of a centralized dispute handling system
- No multi-module tracking (Fines + Payments + Disputes + Reporting)

JURMANA solves these gaps.

3. Problem Definition and Objectives

3.1. Problem statement

There is no unified digital system that allows users to view, verify, and pay traffic fines with automated notifications, secure transactions, and a structured dispute-handling mechanism. Manual operations slow down fine processing and reduce transparency.

3.2. Objectives of the project

- Build a secure cloud-based traffic fine payment system
- Enable online fine lookup and payment
- Provide user-friendly dashboards for admins
- Maintain dispute resolution workflows
- Automate OTP, receipt, and update notifications
- Produce analytical reports for government use

3.3. Expected outcomes

- 100% online fine management
- Reduced processing time for payments
- Transparency in administration
- More efficient revenue collection
- Citizen convenience and satisfaction

4. System Analysis

4.1. Requirements analysis (Functional and Non-functional)

4.1.1 Functional Requirements

4.1.1.1 User Registration & Login

- **Description:** Allows citizens to create an account and log in using mobile number/email
- **Input:** User enters mobile number/email
- **Output:** User successfully logged in or registered.
- **Error:** Invalid/expired OTP → system shows “OTP verification failed”.

4.1.1.2 Fine Issuance by Admin

- **Description:** Admin can issue traffic fines against a vehicle or citizen.
- **Input:** Vehicle number, violation type, fine amount, evidence (images).
- **Output:** Fine is generated and stored; user notified.

- **Error:** "Failed to issue fine. Please fill all fields."

4.1.1.3 View Fines by Vehicle/User ID

- **Description:** Users or admins can view fine details using vehicle number or user ID.
- **Input:** Vehicle number / User ID.
- **Output:** System displays all pending & paid fines.
- **Error:** Invalid ID → "No records found".

4.1.1.4 Online Payment Processing

- **Description:** Enables users to pay fines through online payment gateways.
- **Input:** chooses payment method → proceeds to pay.
- **Output:** Payment success and receipt generated**.**
- **Error:** Payment failure → "Transaction failed. Try again."

4.1.1.5 Store Transaction History

- **Description:** Saves details of all user payments and fines.
- **Input:** Payment confirmation and metadata.
- **Output:** Transaction stored and viewable in history.
- **Error: Database issue → "Unable to save transaction".**

4.1.1.6 Help & FAQ Section

- **Description:** Provides a searchable knowledge base for troubleshooting and general usage guidance.
- **Input:** User enters a query or selects a topic.
- **Output:** Displays relevant help articles or steps.
- **Error:** "No results found" message if topic not available.

4.1.1.7 Generate Dashboards & Reports

- **Description:** Admin can view statistical insights, revenue reports, dispute analytics, etc.
- **Input:** Time range, filters, report type.
- **Output: Visual dashboard graphs & downloadable reports.**
- **Error:** "No report data found."

4.1.2 Non-Functional Requirements

4.1.2.1 Reliability – 99% Uptime

- Description: System should remain operational most of the time without failures.
- **Input:** Normal user activity.
- **Output:** Stable system performance.
- **Error:** Unexpected downtime triggers an alert to admin.

4.1.2.2 Security – Encryption & OTP

- Description: Ensures all user data, passwords, and transactions are secure.
- **Input:** Login, payments, and data entry.

- **Output:** Encrypted storage & secure communication.
- **Error:** Unauthorized access attempt → security alert triggered.

4.1.2.3 Performance – Payment within 5 Seconds

- Description: Payment processing must be fast and responsive.
- **Input:** User initiates payment.
- **Output:** Payment response within 5 seconds.
- **Error:** Timeout → "Payment taking too long. Try again."

4.1.2.4 Usability – Simple UI

- Description: System must be easy for citizens to understand and navigate.
- **Input:** User interactions with UI.
- **Output:** Smooth navigation and clear workflows.
- **Error:** Confusing UI → low usability feedback.

4.1.2.5 Scalability – 5000+ Concurrent Users

- Description: System must handle peak loads efficiently.
- **Input:** Large number of simultaneous users.
- **Output:** Stable performance without slowdown.
- **Error:** Server overload → "Too many users. Please try again."

4.1.2.6 Maintainability – Modular Design

- Description: The system architecture should allow easy updates and debugging.
- **Input:** Code updates/patches.
- **Output:** Changes applied without affecting system functionality.
- **Error:** Bad module update → rollback initiated.

4.1.2.7 Compatibility – Browser-Independent

- Description: System should work smoothly on all major browsers.
- **Input:** Access from Chrome/Firefox/Edge/Safari.
- **Output:** Consistent UI rendering.
- **Error:** Browser incompatibility → fallback to standard version.

4.2. Feasibility study (Technical, Economic, Operational)

1. Technical Feasibility

- Uses modern tech: Node.js/React/MySQL
- Integrates easily with APIs

2. Economic Feasibility

- Medium–scale project
- Dev cost estimated at **₹11,78,500**
- ROI is high due to recurring fine payments

3. Operational Feasibility

- Easy adoption by traffic departments
- Reduces workload
- Users require only mobile number + vehicle details

4.3. Use case diagrams and descriptions

Use case diagrams visually represent how different users (actors) interact with the Online Payroll Management System (OPMS). They help in understanding system behaviour from the end-user perspective and ensure that all functional requirements are properly addressed in the web application.

Use Case	Primary Role in the System
Register / Login	Allows users (citizens) to create an account, verify identity via OTP, and securely log into the system to access services.
View Fines	Enables users to search and view all pending/paid fines using vehicle number or user ID, including violation details and due amounts.
Pay Fine	Allows users to pay fines online through the integrated payment gateway. Confirms payment, generates receipt, and stores transaction history.
Raise Dispute	Users can challenge wrongly issued fines by submitting documents/evidence. The system forwards disputes to admin for verification.
Admin Issue Fine	Administrators can generate new fines by entering violation details, uploading evidence, and assigning the fine to a vehicle/user.
Admin Generate Reports	Administrators can generate dashboards, revenue reports, payment analytics, dispute summaries, and daily operational statistics.

Use Case Diagram Description (Textual Form)

The functionalities of CableSetGo are grouped into logical modules that support smooth operations for plan management, subscription renewal, customer service, and admin-level control. The use cases for each module are described below.

I. User & Subscription Management Module

- **User Registration & Login:** Allows the Customer to create an account, log in securely, and manage their profile.
- **Browse & Select Plans:** Enables users to view available cable/OTT plans, compare features, filter based on price or duration, and choose a suitable subscription.
- Manage Subscriptions : Customers can view active plans, check validity, and renew or upgrade subscriptions.

II. Payment & Recharge Module

- **Online Payment System :** Customer initiates a recharge payment. The Payment Gateway receives the request and returns success/failure status.
- **Invoice Generation :** If payment is successful, the system produces a downloadable PDF invoice and sends notifications via SMS/Email.

III. Complaint & Support Module

- **Raise Complaint:** Customers can submit service complaints, billing issues, or technical queries.
- **Track Complaint Status:** Users can check whether their complaint is *Pending, In-Progress, or Resolved.*
- Admin Complaint Handling: Administrator reviews and updates complaint status.

IV. Plan & Operator Management Module (Admin)

- **Add/Update Plans:** Administrator creates new plans, updates pricing, or modifies available channels/features.
- **Manage Users:** Admin can view, deactivate, or modify customer accounts.
- **Dashboard & Reports**: Admin views subscription statistics, payment summaries, and complaint reports.

V. System Administration & Security Module

- **User Authentication & Authorization**: Secure login for customers and administrators with role-based access control.
- **Backup & Restore Database:** Administrator can generate backups and restore system data if required.
- **Notifications System:** System sends SMS/Email alerts for subscription expiry, successful payments, and complaint updates.

Use Case Summary

The CableSetGo system revolves around three main user flows:

1. Customer

Citizens are the primary users of the system. They:

- Register and log in using OTP authentication
- View pending and paid traffic fines using vehicle number
- Make secure online payments
- Download receipts and check transaction history
- Raise disputes by submitting documents or evidence

- Track dispute resolution status
- Receive SMS/Email notifications

They use the system mainly for **self-service fine management**, making the process faster and more transparent.

2. Administrator

Administrators oversee system operations and ensure accuracy and fairness. They:

- Issue traffic fines with violation details and evidence
- Manage user accounts and vehicle records
- Verify and resolve disputes raised by citizens
- Generate reports and dashboards (revenue, violations, disputes)
- Monitor payment status and system activity
- Maintain data integrity and handle system-level controls

Admins ensure the enforcement system remains **efficient, accurate, and well-managed**.

3. Payment Gateway

The Payment Gateway is an external service integrated into the system. It:

- Processes online fine payments securely
- Verifies transaction details
- Returns success or failure status for each payment
- Ensures encrypted communication during the transaction

This component ensures **safe and reliable online payments** for citizens.

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5. System Design

5.1. System architecture diagram

Presentation Layer (Frontend)

The Presentation Layer is the interface through which **Citizens (Users)** and **Administrators** interact with the *Online Traffic Fine Payment System (Jurmana)*.

Developed using **HTML, CSS, JavaScript**, and optionally frameworks like **React/Bootstrap**, this layer allows users to:

- Register/Login (OTP-based)
- View issued fines
- Pay fines online
- Raise disputes with documents
- Receive payment status and notifications
- View transaction history

Admins use this layer to:

- Issue fines
- Resolve disputes
- Generate reports and dashboards

The frontend performs basic form validation and communicates with the backend through **secure HTTPS API calls**.

It is fully responsive, browser-independent, and designed to ensure a simple and intuitive user experience for citizens.

Application Layer (Business Logic / Backend)

The Application Layer contains the core logic of the traffic fine system and is implemented using **Node.js / PHP / Java (depending on your project stack)**. This layer handles all system operations, such as:

- OTP-based authentication
- Fine creation & validation
- Online payment processing
- Dispute submission and review
- Notification triggers (SMS/Email)
- Admin actions (manage fines, disputes, reports)

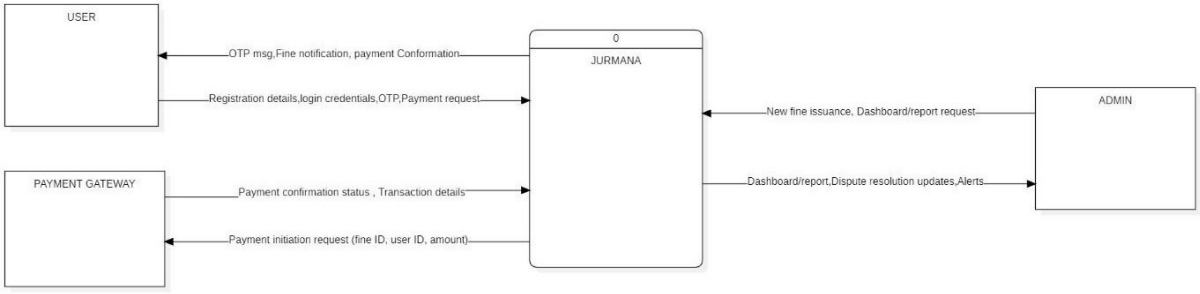
The backend enforces **role-based access control**, ensuring secure operations by both users and administrators.
It passes data between the frontend and database and also integrates with **external Payment Gateways** for secure fine payment processing.
All actions like fine issuance, dispute resolution, and payments follow predefined business rules for consistency and security.

Database Layer (Data Storage & Management)

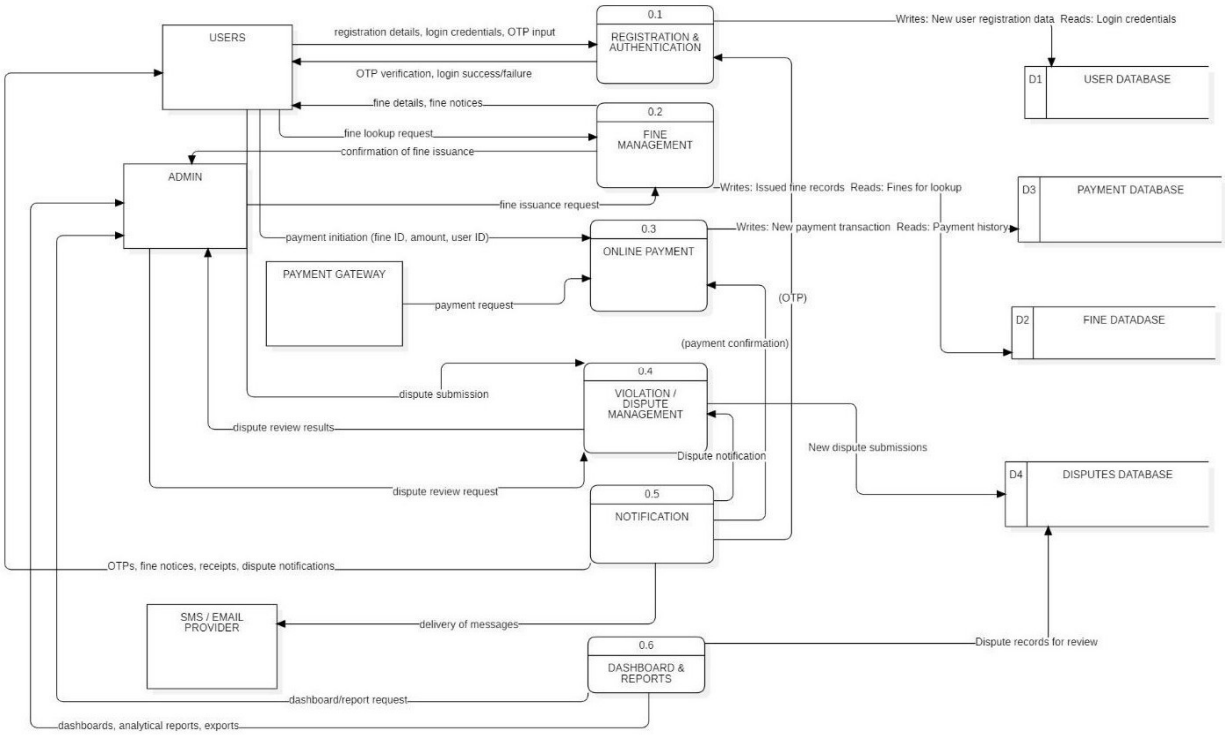
- This layer uses **MySQL** to store all essential system data, including:
- User accounts & profile details
- Vehicle information
- Issued fines
- Payment records & transaction logs
- Disputes and attached documents
- Admin decisions
- Notification logs
- Dashboard statistics
- MySQL ensures secure, structured, and scalable storage.
  The relational structure allows quick lookups for fine history, payment verification, report generation, and dispute tracking.

5.2. Data flow diagrams (DFD Level 0, Level 1)

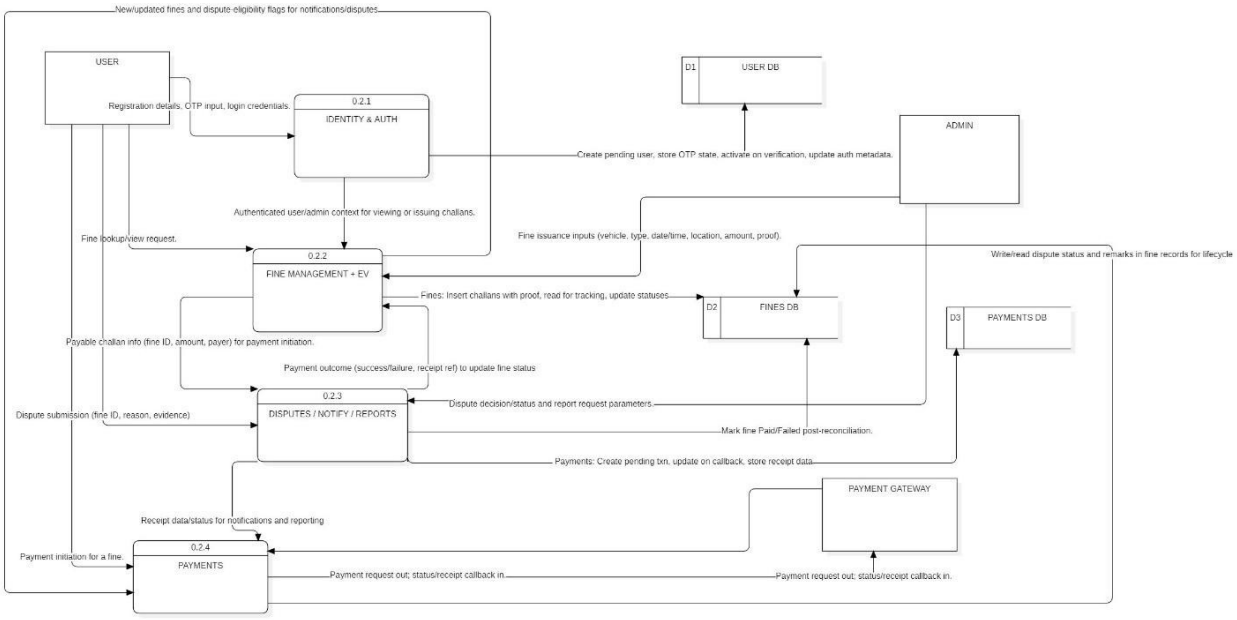
5.2.1 Level 0:



5.2.2 Level 1:



5.2.3 Level 2:



6. Implementation

6.1. Tools and Technologies used

Frontend Technologies

- **Frontend:** HTML, CSS, JavaScript, React
- **Backend:** Node.js / Express
- **Database:** MySQL
- **Other:** REST API, JWT, Nodemailer, SMS Gateway

Bootstrap (optional): Provides responsive layouts and UI components for better design.

Backend Technologies

- **Screenshots from your project UI (login, dashboard, payment page)**
- **Backend includes:**
- **Auth APIs**
- **Fine APIs**
- **Payment handling**
- **Dispute management**

Database Technologies

- **MySQL:** Stores customer accounts, plan details, subscription history, payment transactions, complaints, and notifications in a structured format.
- **phpMyAdmin:** Used for managing tables, running SQL queries, and visualizing CableSet-Go's database structure.

Modeling & Documentation Tools

- **StarUML / Draw.io:** Used for UML diagrams (Use Case, Class, Sequence, Activity).
- **MS Word / Google Docs:** For preparing documentation and reports.

Code Editor / IDE

- **Visual Studio Code / Sublime Text:** Used for writing and editing HTML, CSS, JavaScript, and PHP code.

Version Control (Optional)

- **Git / GitHub:** Helps track changes, maintain project versions, and store the entire codebase safely for future improvements.

6.2. Frontend and Backend description

Frontend Description

 Screenshots from your project UI (login, dashboard, payment page) Backend
 includes:

- Auth APIs
- Fine APIs
- Payment handling
- Dispute management

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Backend Description

The backend forms the core processing engine of **OTFPS**, developed using **PHP** and hosted on an Apache server through XAMPP. It manages all system logic, secure access control, and communication with the MySQL database.

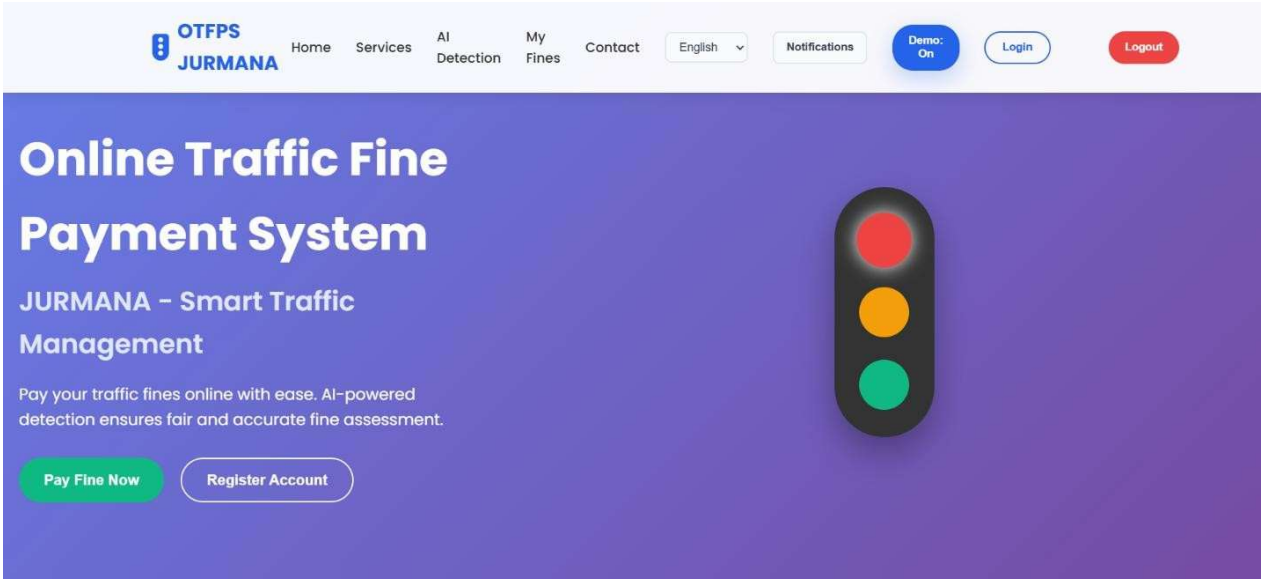
Key backend responsibilities include:

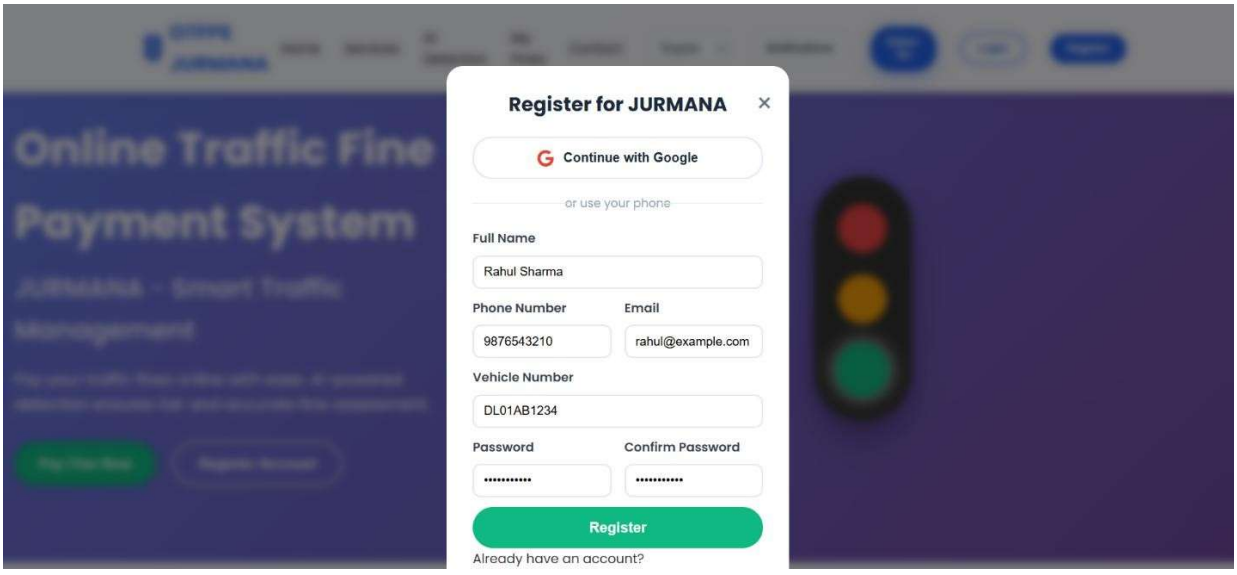
- User authentication and secure login
- Role-based access for Administrator and Customer
- Managing user and admin accounts
- Handling plan retrieval and updates
- Processing subscription renewals and sending payment requests
- Integrating with the Payment Gateway and verifying transaction responses
- Generating invoices and sending email/SMS notifications
- Recording and updating complaints
- Storing transaction logs and subscription history
- Preparing admin reports (plan usage, revenue, complaints)

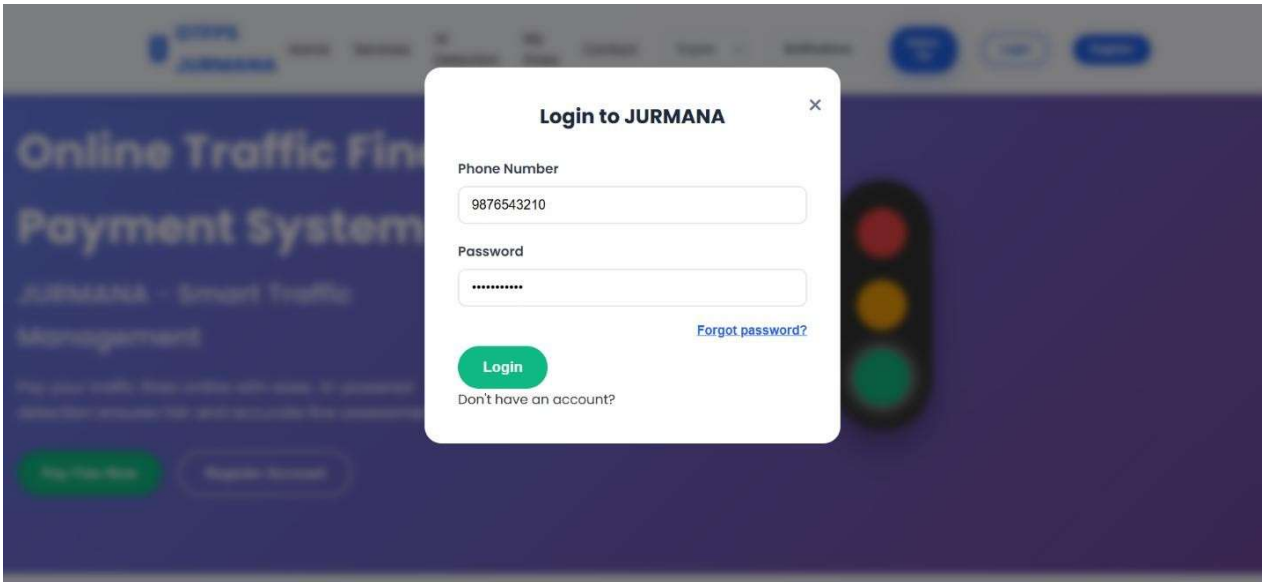
The backend interacts with the **MySQL** database using SQL queries to ensure that payroll data is stored, retrieved, and processed efficiently.

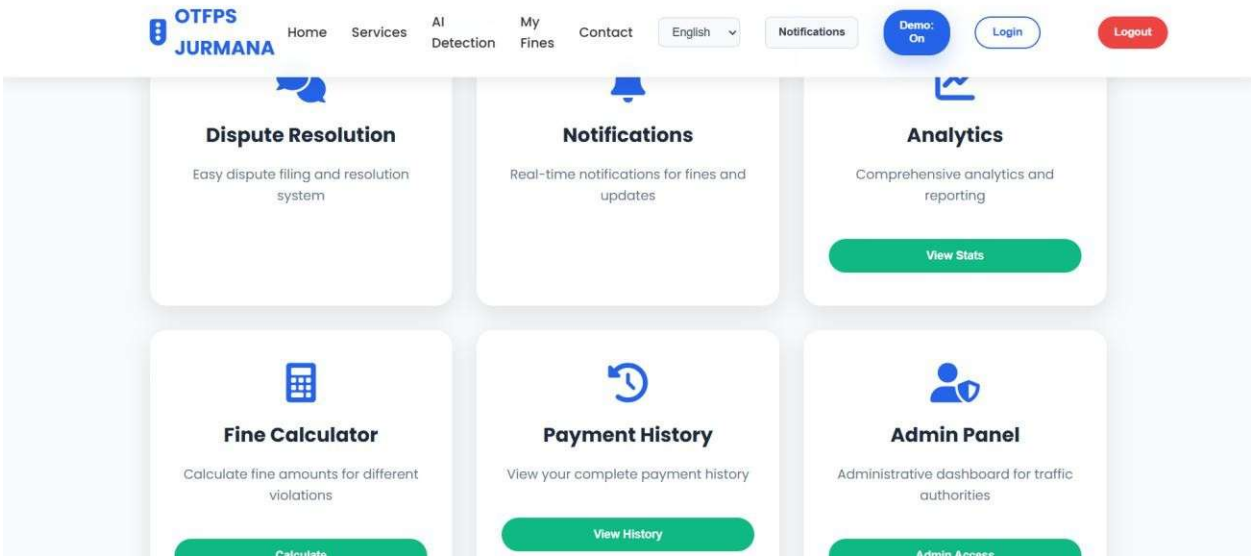
7. Results and Discussions

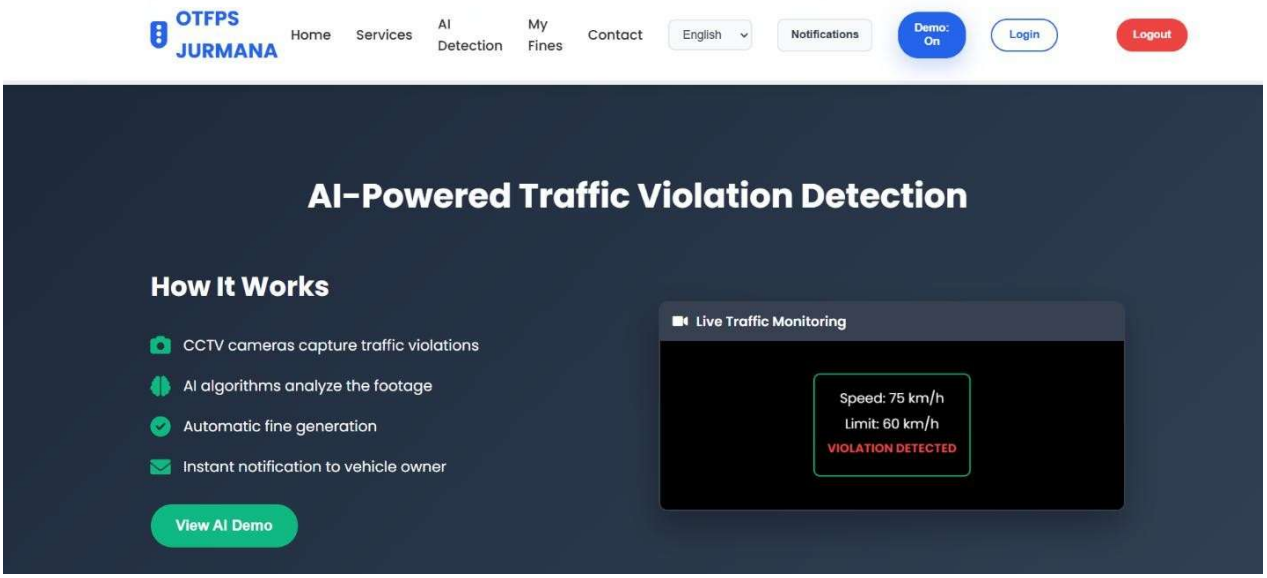
7.1. Screenshots of the application / Output demonstration

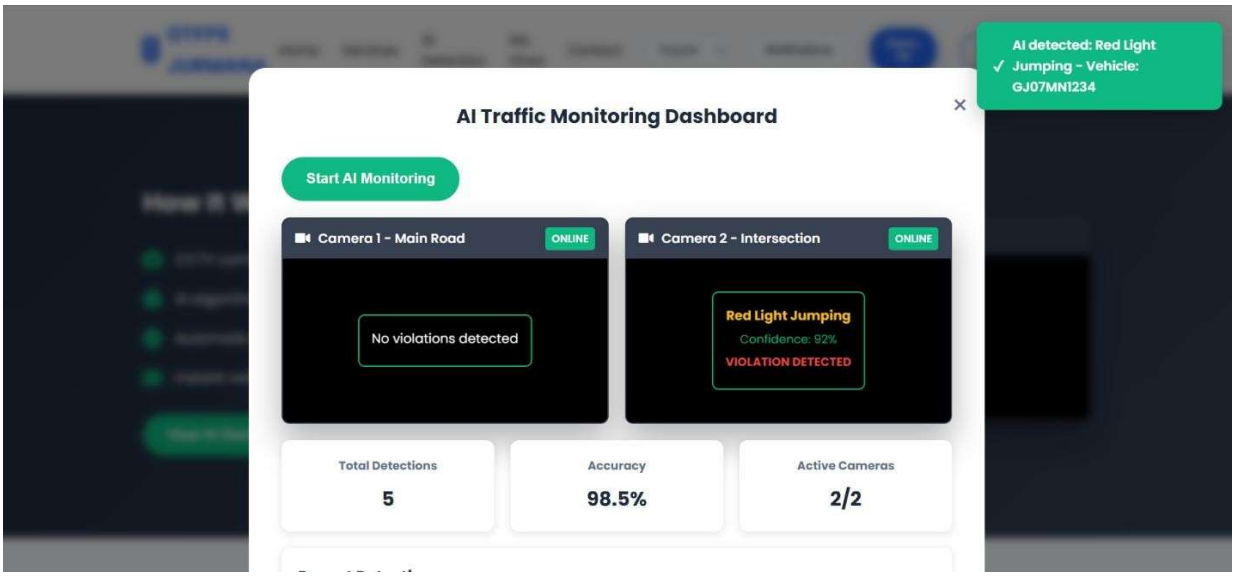


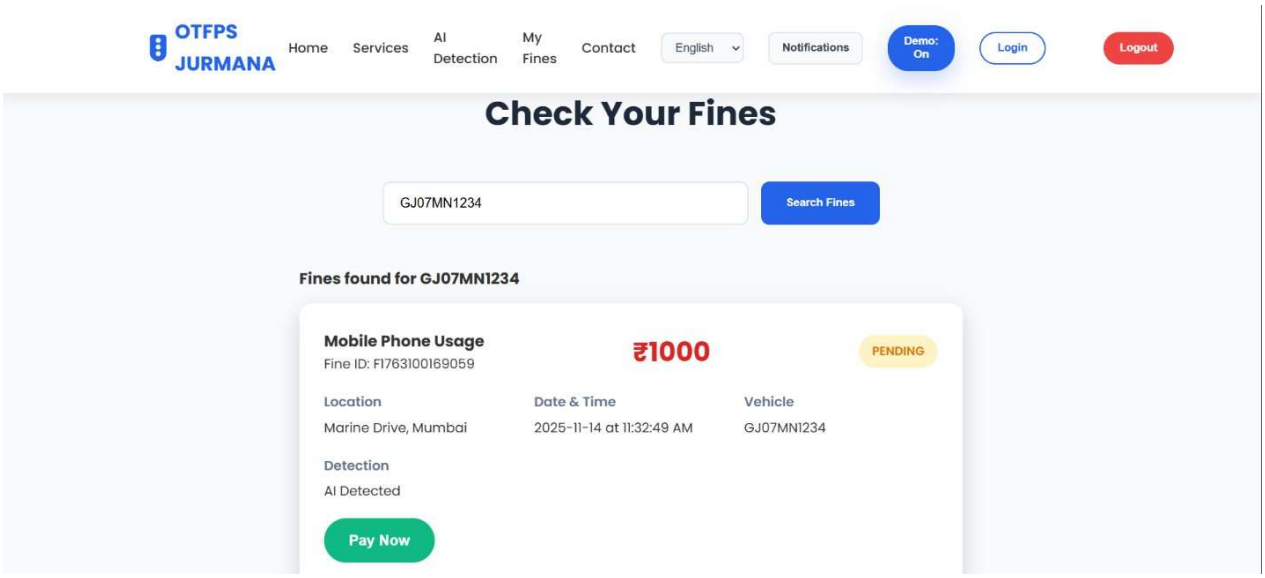


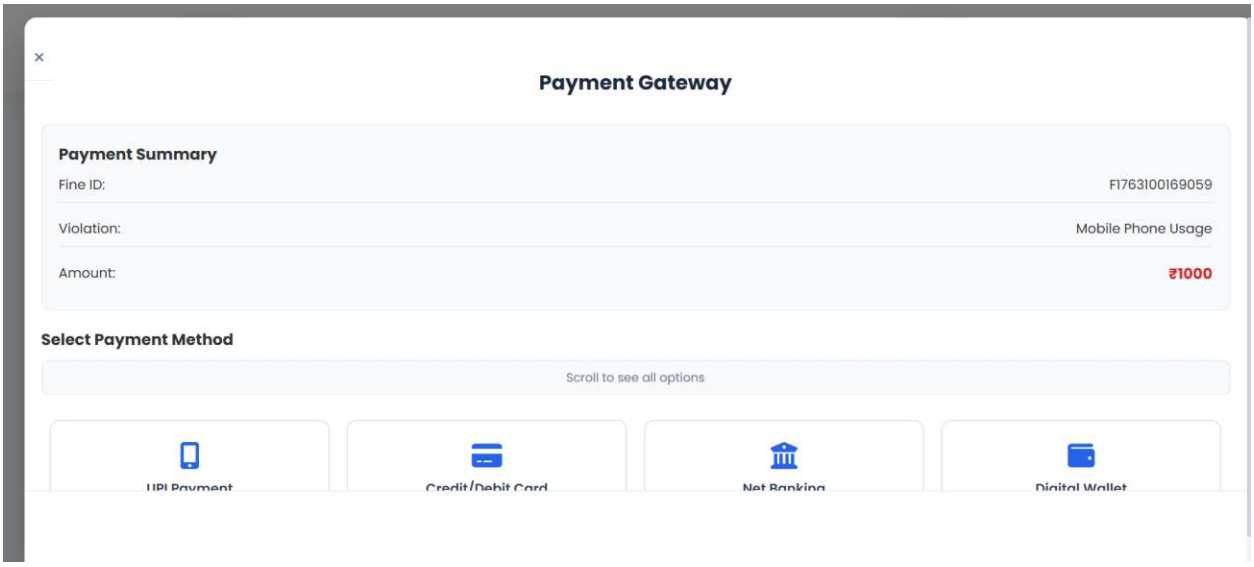

Login to JURMANA
Phone Number
9876543210
Password
Forgot password?
Login
Don't have an account?

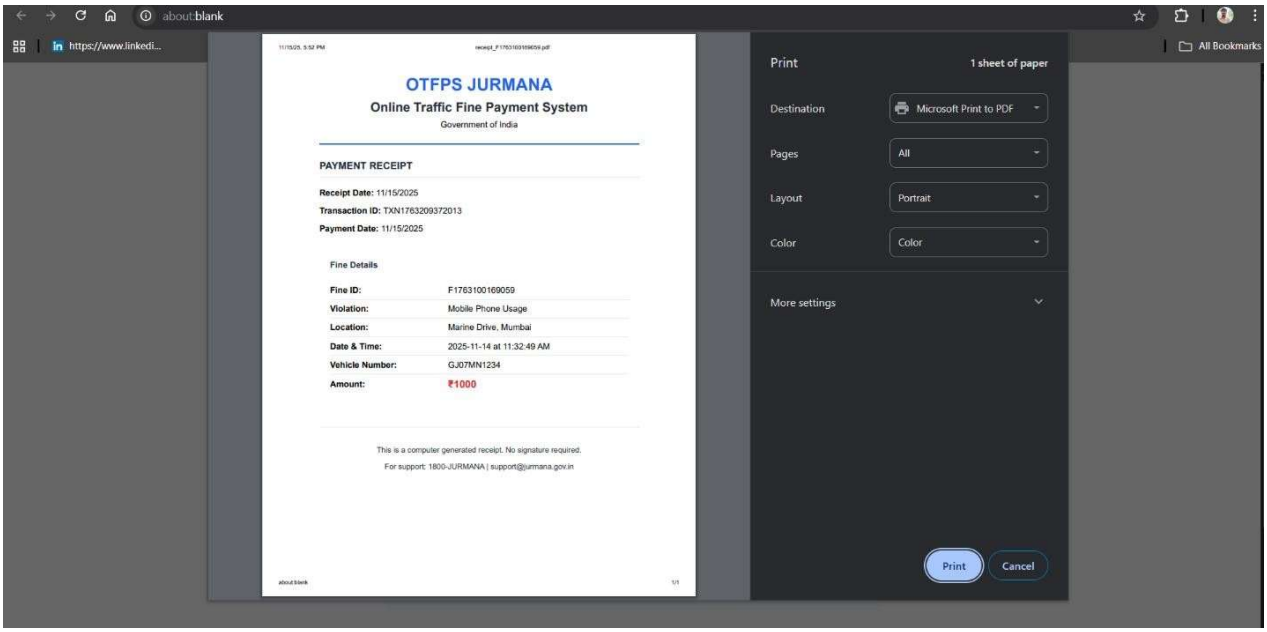

OTFPS JURMANA
Home
Services
AI Detection
My Fines
Contact
English
Notifications
Demo: On
Login
Logout
Dispute Resolution
Easy dispute filing and resolution system
Notifications
Real-time notifications for fines and updates
Analytics
Comprehensive analytics and reporting
View Stats
Fine Calculator
Calculate fine amounts for different violations
Payment History
View your complete payment history
View History
Admin Panel
Administrative dashboard for traffic authorities


OTFPS JURMANA
Home
Services
AI Detection
My Fines
Contact
English
Notifications
Demo: On
Login
Logout
AI-Powered Traffic Violation Detection
How It Works
CCTV cameras capture traffic violations
AI algorithms analyze the footage
Automatic fine generation
Instant notification to vehicle owner
View AI Demo
Live Traffic Monitoring
Speed: 75 km/h
Limit: 60 km/h
VIOLATION DETECTED


AI detected: Red Light Jumping - Vehicle: GJ07MN1234
AI Traffic Monitoring Dashboard
Start AI Monitoring
Camera 1 - Main Road
ONLINE
No violations detected
Camera 2 - Intersection
ONLINE
Red Light Jumping
Confidence: 92%
VIOLATION DETECTED
Total Detections
5
Accuracy
98.5%
Active Cameras
2/2


OTFPS JURMANA
Home
Services
AI Detection
My Fines
Contact
English
Notifications
Demo: On
Login
Logout
Check Your Fines
GJ07MN1234
Search Fines
Fines found for GJ07MN1234
Mobile Phone Usage
Fine ID: FI763100169059
₹1000
PENDING
Location
Marine Drive, Mumbai
Date & Time
2025-11-14 at 11:32:49 AM
Vehicle
GJ07MN1234
Detection
AI Detected
Pay Now


Payment Gateway
Payment Summary
Fine ID:
FI763100169059
Violation:
Mobile Phone Usage
Amount:
₹1000
Select Payment Method
Scroll to see all options
UPI Payment
Credit/Debit Card
Net Banking
Digital Wallet



7.2. Analysis of results

Citizens access the system to perform all fine-related activities.
They can register via OTP, log in securely, view fines issued against their vehicle, make online payments, download receipts, and raise disputes by uploading supporting documents.
They also receive updates through SMS/Email notifications regarding fine issuance, payment success, or dispute outcomes.
Overall, users primarily rely on the platform for **self-service fine management** without visiting physical offices.

Administrators handle all official operations within the system.
They issue new fines, upload violation evidence, update fine details, verify documents, resolve disputes, and generate dashboards and report summaries for monitoring revenues and violations. They also maintain user records, oversee payment reports, ensure system security, and manage routine backend tasks.
Admins ensure the platform remains accurate, transparent, and up-to-date.

The Payment Gateway securely processes online traffic fine payments.
It handles digital transactions initiated by citizens, verifies payment information, and returns success or failure responses instantly.
It helps maintain safe, encrypted, and reliable financial operations within the platform

Smart and Real-Time Location Assistance
The **Find Nearest Traffic Office / Police Station module** enhances user experience by:

- Automatically detecting the user's live location (with permission)
- Locating the nearest authorized **Traffic Department Branch / Police Control Room** within a configurable radius
- Displaying complete branch details such as:
    - Office name
    - Contact number
    - Working hours
    - Officer-in-charge
- Providing a guided route via map redirection (Google Maps)

This feature reduces citizen confusion, helps them visit the correct office for verification or dispute support, and enables **faster issue resolution**.

Organized Plan Selection and Comparison
The system displays **traffic fines** in a well-structured card layout that includes:

- Violation description
- Fine amount
- Evidence/proof upload (image/video by authority)
- Date & time of violation
- Vehicle details
- Due date and penalty escalation (if any)
- Status (Unpaid / Paid / Under Dispute)

The organized layout helps citizens understand the violation instantly, reducing confusion and ensuring faster decision-making for payment or dispute submission.

Instant Digital Receipt Generation
After successful payment, users automatically receive a digital receipt containing:

- Receipt Number / Transaction ID
- Payment date & timestamp
- Citizen details
- Vehicle number
- Fine ID & violation details
- Amount paid
- Payment method
- Issuing authority
- Verification seal / QR code

This enhances transparency, improves trust, and eliminates the need for manual verification at traffic offices.

Conclusion of Analysis
Overall, the **OTFPS – JURMANA platform** demonstrates how a modern web-based system can significantly improve:

- Access to fine details
- Real-time notifications
- Payment processing
- Dispute resolution
- Office/branch location guidance

The results clearly show that the platform meets its core objectives through:

- Faster user workflows
- Organized fine management
- Transparent payment and receipt handling
- Smart location-based office assistance
- Improved professionalism and trust

This project delivers a **practical, real-world-ready solution** for digital traffic fine management.

8. Conclusion

8.1. Summary of achievements

The **Online Traffic Fine Payment System (OTFPS – JURMANA)** successfully delivers a smooth and accessible platform for citizens to:

- View pending traffic fines
- Access violation details with proof
- Log in securely (OTP verification)
- Pay fines online instantly
- Locate nearby traffic offices
- Raise disputes with evidence
- Receive digital receipts

The project achieves key objectives including:

- Enhanced user experience
- Clear violation details and fine structure
- Transparent payment summaries
- Automated notifications (OTP, payment receipt, dispute updates)
- Modern and professional UI suitable for large-scale deployment

8.2. Limitations

The current version is demonstration-ready but not fully production-ready due to:

- Use of demo/static data instead of a live backend
- No real database for storing fines, transactions, or user history
- Location detection and office mapping simulated
- No real payment gateway (Razorpay/UPI) integrated yet

- Security features like encryption, role-based authorization, and audit logs not fully implemented

9. Future Scope

9.1. Future enhancements

The OTFPS platform provides a strong starting point, and several enhancements can make it production-ready:

1. Full Backend Integration

Implementing a real backend (Node.js, Django, Laravel) with a relational database (MySQL/PostgreSQL) for:

- Permanent fine records
- Payment history
- Dispute workflows
- Admin dashboard data

2. Live Payment Gateway Integration

Using APIs such as:

- Razorpay
- Paytm
- Stripe
- UPI Intent Payments

This will enable real transactions and enhance authenticity.

3. Real-Time Location Assistance Using Google Maps API

Implement:

- Accurate distance calculation
- Dynamic branch listings
- Geo-fencing for identifying regional jurisdiction
- Directions to nearest police station or traffic office

4. Mobile Application (Android/iOS)

To provide:

- Instant fine notifications
- UPI-based quick payments
- Dispute tracking
- Location guidance

5. Admin Analytics Dashboard

Reports for authorities such as:

- Total fines collected
- Unpaid vs paid fines
- Most common violations
- Citizens' dispute patterns
- Regional fine distribution heatmap

This will support better government decision-making.

10. References

10.1. Github Repository Link:

- R. Sinha, *JURMANA – Online Traffic Fine Payment System (OTFPS)*, GitHub Repository.
  Available at: **https://github.com/ragyam-sinha/jurmana_8558.git**

10.2. OPMS Website:

- Live Deployment of JURMANA – Online Traffic Fine Payment System
  Available at: **https://jurmana-8558.vercel.app/**

Case Study References

- **MDN Web Docs – HTML, CSS, JavaScript**
  Reference for UI and frontend behavior
  https://developer.mozilla.org/

- **Supabase / Firebase Documentation (for demo authentication concepts)**
  https://supabase.com/docs

- **Google Maps Geolocation API**
  Used for designing location-based office detection
  https://developers.google.com/maps/documentation

- **W3C Web Accessibility Guidelines**
  https://www.w3.org/WAI/standards-guidelines

- **Bootstrap & UI/UX Design Guidelines**
  Reference for design patterns and interface clarity

- **B. Boehm, Software Engineering Economics**
  Used during COCOMO estimation

- **Pressman, Software Engineering Approach**
  Reference for SDLC and planning models

11. Appendix – add acronyms and code

Acronym	Full Form
OTFPS	Online Traffic Fine Payment System

Acronym	Full Form
DBMS	Database Management System
OTP	One-Time Password
API	Application Programming Interface
GPS	Global Positioning System
UI	User Interface
UX	User Experience
JSON	JavaScript Object Notation
SQL	Structured Query Language
UPI	Unified Payments Interface
HTTP	HyperText Transfer Protocol
HTTPS	HyperText Transfer Protocol Secure
CDN	Content Delivery Network
DOM	Document Object Model
ISP	Internet Service Provider *(not required but optional)*